

Acute Lymphoblastic Leukemia Detection Employing Deep Learning and Transfer Learning Techniques

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Abstract- The death due to cancer increases daily due to late or even no diagnosis. It is necessary for early detection and treatment of such fatal cancers and avoiding carcinogenic agents is an important step for the prevention of cancers. The most frequent kind of cancer, with prostate and lung cancer following closely after. Different types of cancers are detected by collecting blood or urine samples. The majority of the cancers are detected by Computed Tomography (CT), Magnetic Resonance Imaging (MRI), and even X-Ray images. There are different techniques used by the doctors like pattern recognition, seven-point detection, Asymmetry, Border, Color, Diameter (ABCD) method. There are lot many methods used by professionals in this field. However, these methods may not be efficient always. Hence currently there is a huge scope for automation in the medical field for better performance and early diagnosis of fatal diseases to prevent further complexities and also for the betterment of mankind. This paper demonstrates the automation of the categorization of Acute Lymphoblastic Leukaemia. In order to do this, many CNN variations with various activation functions are tested; CNN, Xception, and MobileNetV2 yielded the best accuracy of 100%. These automated diagnostic approaches reduce the risk of cancer in humans because they are more accurate than manual diagnosis methods.

Keywords- ALL (Acute Lymphoblastic Leukemia), ABCD (Asymmetry, Border, Color, Diameter), CNN (Convolution Neural Network)

I. INTRODUCTION

Plasma and well-known cells—white blood, red blood, and platelets—make up the blood, each of which has a specific function. RBCs are responsible for transporting oxygen throughout the human system via the pulmonary system. WBC aid the immune system in combating foreign invaders like viruses and bacteria. Clotting platelets aid in stopping bleeding [1-2]. Leukemia is a kind of blood cancer identified by the presence of an aberrant high number of WBCs, that remain young and may cause

problems for the increase in the number of RBCs and platelets your body produces. This is how the human physique's WBC to RBC ratio is 1:1,000. This ratio indicates that there is just one white blood cell for every thousand red blood cells [3-4]. Lymphocytic or lymphoblastic leukemia is the kind of leukemia that originates in lymphoid cells, whereas myelogenous or myeloid leukemia originates in myeloid cells. According on how quickly the cells are dividing, leukemia may be classified as either acute or chronic. In acute leukemia, the aberrant blood cells are often dysfunctional blasts (young cells). There is rapid expansion of these cells. Without urgent treatment, acute leukemia rapidly worsens. Chronic leukemia not only produces immature blood cells but also mature, fully functional cells. In chronic leukemia, the proliferation of blasts is sluggish. The progression of the illness is slowed [5]

ALL, often referred to as blood cancer, is a kind of childhood cancer that affects the bone marrow and blood that accounts for 25% of pediatric cancer cases. [6-7]. This condition manifests as discomfort in the bones, gum bleeding, high temperature, increased infection rates, serious or ongoing nasal bleeding, dyspnea, weaknesses, or exhaustion [8]. An expert examines this by taking blood samples, and a classification-based conclusion is made. DL is essential for automating the process of classifying and detecting ALL cells [8a]. The image of normal cells is described in Fig.1 (a) and infected leukemia cells are shown in Fig.1 (b).

Since the illness may be lethal if neglected, immediate diagnosis and therapy are crucial, as is accurate detection. In ALL, the survival rate is 68.8%. Radioactive exposure, certain hereditary conditions, and a prior history of cancer therapies all raise the chance of ALL. The immature cells created during this process are referred to as lymphoblasts. Radiation therapy, chemotherapy, bone

marrow transplantation, and several other procedures are used to treat this illness [9–11].

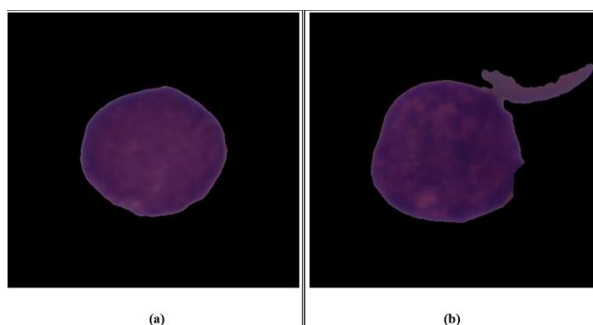


Fig. 1 (a). A normal cell is identified by its circular shape and complete cell wall covering. (b). Leukemia Blast is identified by its abnormal cell structure and broken cell wall covering [8a].

Deep learning is used to identify the present remedies to issues that take a long time for experts to resolve. These days, machine learning methods are being used to identify the different illnesses that are circulating the globe [12–13]. Various traditional ML and DL methods are employed to various clinical diagnosis, including sepsis [20], respiratory infections [14–15], and many more. The identification of celestial beings in astronomy is made possible by the extension of DL methods [16–18].

II. LITERATURE SURVEY

This section encompasses a range of pertinent research works.

The work in [1] utilized a shallow CNN for image adaptation and a deep CNN for classification on ALL-IDB2, yielding 96.8% accuracy for unsharp original images. Their study introduced inventive and adapt-able image processing techniques. In [2], the work presents the heterogeneous ALL-IDB2 dataset, containing 1,700 cancerous blood cell images, was employed. Various CNN variants were used for data pretraining, achieving an optimal accuracy of 98% using DenseNet121. Future work aims to further enhance test performance. [3] introduced the LeuFeatx model, attaining a 96% accuracy rate with the ALL-IDB2 dataset. [4] proposed a hybrid model merging properties, achieving the highest accuracy of 99.39%. Their findings revealed areas for future implementation as accuracy increased but retained consistency between training and testing data. [5] introduced a DL-based automated technique for leukemia detection, achieving nearly 100% accuracy across ALLDB1, ALLDB2, and ASH databases. Such efficient models inspire researchers to apply similar strategies across different databases and diseases. [6] conducted a literature review on transfer learning techniques for leukemia differentiation, noting accuracies ranging from 75% to 95% in major studies [7–9].

III. PROPOSED METHODOLOGY

The above work presents a comprehensive examination of various CNN variants, utilizing the Adam optimizer along with ReLU techniques. The method flowchart is displayed in Fig.2., outlines the step-by-step workflow followed in this section. The process begins with data pre-processing and augmentation, expertly handled by the ImageDataGenerator. The dataset is then divided into training and testing sets. CNN are among the learning models in DL that have been constructed designed for leukemia image classification. These models are meticulously demonstrated utilizing different activation functions. A thorough comparative study is undertaken to determine which model is the most efficient activation function employed in this section. This comparison serves as a valuable reference for readers, encouraging them to explore models with diverse activation functions. The models undergo rigorous verification, and the final outcome is a user-friendly web application developed to showcase the best-performing model

The main objective of this work is to explore different variants of CNN, along with various optimizers and activation functions, for analyzing ALL cell images. ALL is a prevalent form of cancer, and its definitive diagnosis typically requires invasive, expensive, and time-consuming tests. In this, the use of peripheral blood smear (PBS) images becomes crucial for initial cancer screening and identifying tumors from the general population. Yet, owing to the vague character of ALL signs, which can give rise to possible errors in diagnosis, the assessment of these PBS pictures by investigators is difficult. The dataset utilized in this investigation includes 3,256 PBS pictures from 89 patients with a possible case of ALL. The photos were processed and colored by experts in a testing facility at Tehran, Iran's Taleqani Hospital. Hematogones belong under the benign category, whereas ALL, which has three distinct types of malignant lymphoblasts (Early Pre-B ALL, Pre-B ALL, and Pro-B ALL), makes up the malignant category. Images evaluated were captured using a Zeiss camera that was mounted to a microscope, and the magnifier setting was set to 100x. From Fig.3., It is clear from the dataset that used contains approximately 68% ALL cell images and 32% normal cells.

This study focuses on showcasing different variants of CNN combined. Additionally, as part of the project, a web application is developed using Flask. The web app allows users to interact with the CNN models and activation functions, providing an intuitive interface for exploring and understanding their effects on the classification process.

Acquisition of Images: All images/pictures were captured by use of a Zeiss camera. These acquired images serve as input to the pre-processing stage for further analysis and

preparation [19].

Pre-Processing: Pre-processing is a crucial step as it enhances the image data by improving essential image features required for subsequent processing. In this stage, the images are converted to gray scale, and a median filter is applied to remove noise effectively. The removal of noise is necessary to ensure the images are prepared optimally for further processing tasks [20-21].

Segmentation: Segmentation involves the division of an image into multiple distinct segments. The primary objective of segmentation is to transform the image representation into a format that is more convenient for analysis purposes [22-23].

Feature Extraction: It is an important step in handling large and redundant input data for algorithms. It consists of changing the initial taken information into a reduced and representative set of features known as a feature vector. During feature extraction, essential image features necessary for image classification are extracted from the input data. To capture and depict the distinct texture aspects of segmented chest X-ray pictures, texture features are recovered from the segmented image.

Classification: The image/picture can be classified into following categories: Benign, Malignant Lymphoblastic Early Pre-B, Malignant Lymphoblastic Pre-B, or Malignant Lymphoblastic Pro-B.

The Multi-Layer Perceptron (MLP) consists of neurons with non-linear activation functions, making it a powerful tool for image classification. By utilizing the mentioned modules, we can effectively detect the presence of Acute-Lymphoblastic-Leukemia (ALL) in the given image.

When opposed to healthy cells, the malignant lymphoblast cells in acute lymphoblastic Leukaemia often have unusual adhesion and spreading abilities. A possible component of ALL cells' capacity to infiltrate organs and metastasize is their heightened spreading and motility.

On the flip side, research has shown that ALL cells may have reduced spreading and adhesion ability under certain conditions. The kind of ALL, genetic abnormalities, the production of adhesion molecule genes, and the surrounding environment are among variables that might affect the extent and features of cell metastasis.

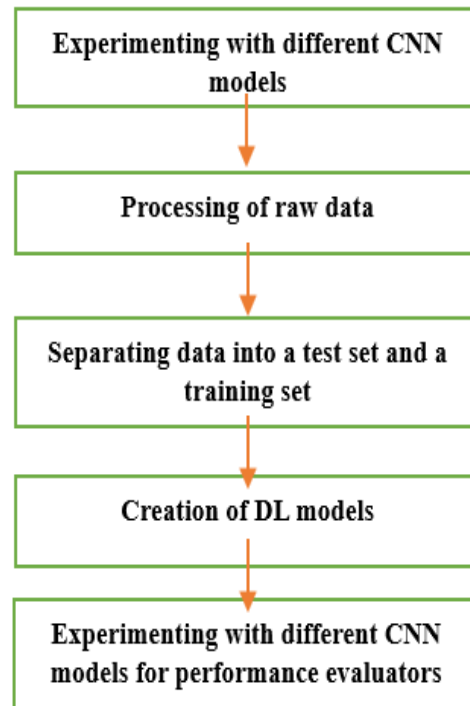


Fig. 2 Block Diagram

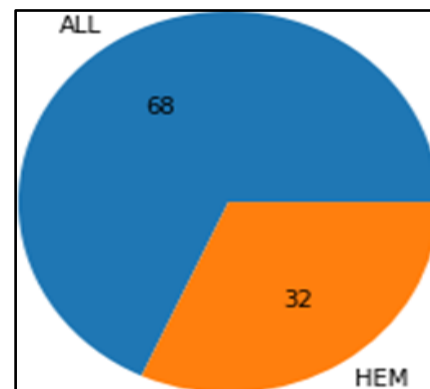


Fig. 3 Spread of ALL and Normal Cells.

IV. RESULTS AND DISCUSSION

A. CNN Model Results:

The accuracy and loss of the Model which is optimized CNN in the Training and Testing has been shown in Fig.4. and Fig.5. Confusion Matrix and Classification Report of Model which is optimized CNN has been shown in Fig.6. and Fig.7.

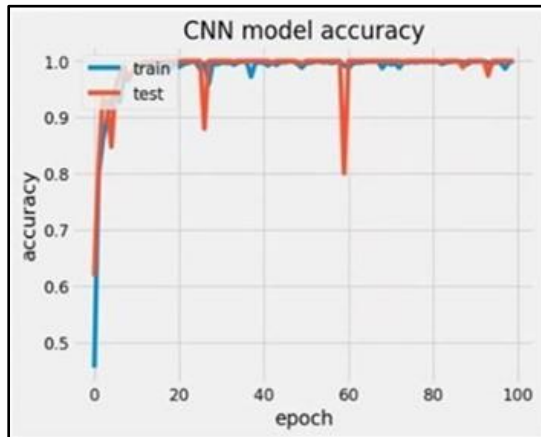


Fig. 4 CNN Model accuracy

The accuracy of a ML model is a measure of its performance on test data. The proportion of photos that the model properly identified is known as accuracy in tasks such as image classification as shown in Fig.4.

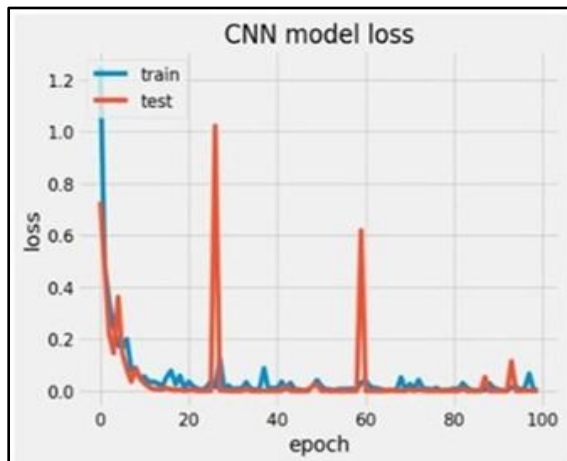


Fig. 5 CNN Model losses

The convolutional neural network (CNN) model trained on data from abnormal, sick cells has its own loss curve, whereas the CNN model trained on data from healthy, normal cells has its own. On one side, we have the training iteration or epoch number, and on the other, we have the loss value (loss).

The CNN model learns discriminative characteristics from both the ALL cells and normal cells more easily, as shown in the comparison of the two loss curves as shown in Fig.5 and Fig.6.

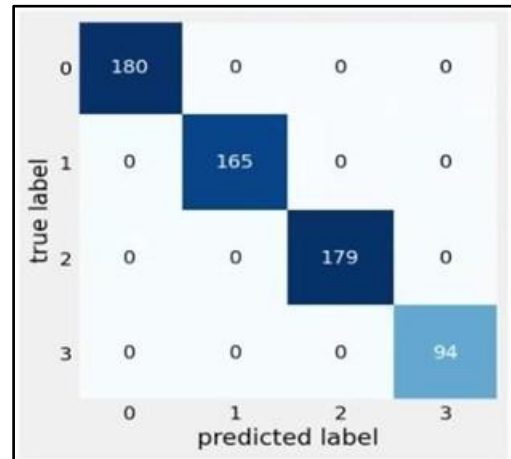


Fig. 6 CNN Model losses

The Fig.7. shows the precision, recall, F1-score and support for the ALL and normal cells classification.

```
# Print the report
print(report)
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	precision	recall	f1-score	support
0	1.00	1.00	1.00	180
1	1.00	1.00	1.00	165
2	1.00	1.00	1.00	179
3	1.00	1.00	1.00	94
accuracy			1.00	618
macro avg	1.00	1.00	1.00	618
weighted avg	1.00	1.00	1.00	618

Fig. 7 CNN Classification Report

B. ResNet50 Model Results:

The accuracy and loss of the ResNet50 Model in the Training and Testing has been shown in Fig.8. and Fig.9. Confusion Matrix of the Res Net50 Model has been shown in Fig.10.

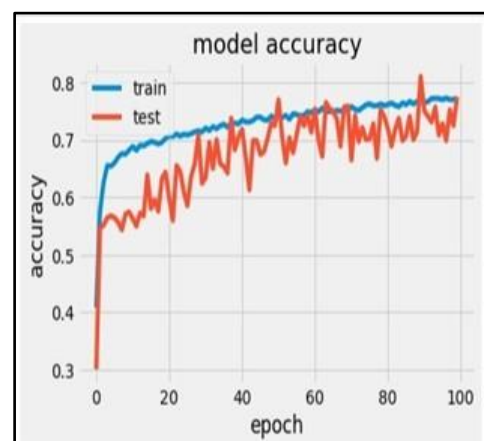


Fig. 8 ResNet50 Model accuracy

Model accuracy curves for training (blue) and test (orange) data across 100 epochs are shown in Fig.8. The test accuracy trails behind with a noticeable gap, suggesting overfitting difficulties, but the training accuracy rises quickly and reaches over 0.7 within 20 epochs, showing successful learning on the training data. Significant oscillations and volatility are seen in both curves, suggesting instability and trouble with a smooth convergence. By the last epochs, the test accuracy is only around 0.65 and the training accuracy is about 0.75, with a persisting disparity suggesting that generalization performance may still be improved.

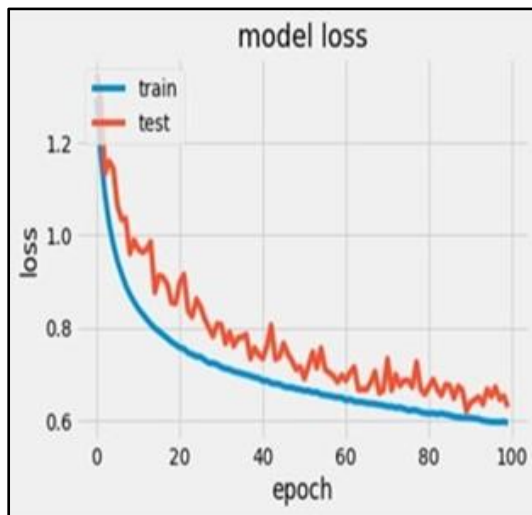


Fig. 9 ResNet50 Model losses

In the early epochs, both test and training loss drop very quickly from their extremely high starting points (around 1.2), suggesting that the model is successfully gaining knowledge from data as shown in Fig.9. It is to be anticipated that early on, the training loss drops considerably quicker than the test loss since the model fits the training data before generalizing. The test loss curve is regularly greater than the training curve, and this difference is clearly obvious. This discrepancy suggests a certain level of overfitting, a situation in which the model fits the training set too closely while underfitting the test set. Particularly in later epochs, both curves exhibit noticeable oscillations and are quite volatile. This volatility points to the possibility of instability or trouble reaching an ideal convergence. The test loss is still around 0.8 by the last few epochs (roughly 100), but the training loss has dropped to about 0.5. This discrepancy suggests that generalization may yet be improved.

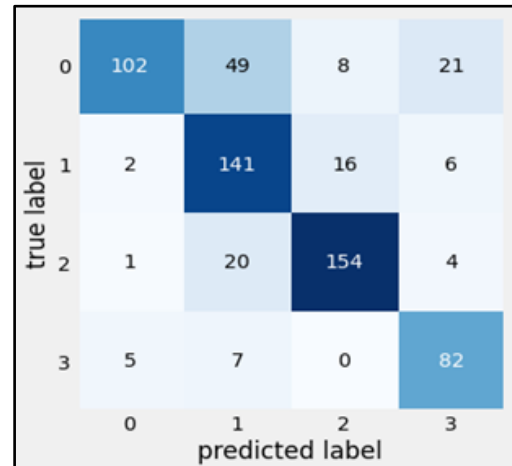


Fig. 10 ResNet50 Confusion Matrix

Similarly, this work exhibits with other algorithms as well and comparison of performance with algorithms are show in below Table I

TABLE 1 EXAMINING THE ACCURACY OF VARIOUS APPROACHES IN CONJUNCTION WITH CLASSIFICATION REPORT

Model	Accuracy	Precision	Recall	F1 Score
CNN	100%	100%	100%	100%
ResNet50	77.50%	80%	78%	77%
Xception	100%	100%	100%	100%
MobileNetV2	100%	100%	100%	100%
VGG19	96.12%	96%	96%	96%

V.CONCLUSION

Acute-Lymphoblastic-Leukaemia (ALL) is a serious illness that is still responsible for a great number of fatalities, and its prevalence is on the rise. One potential field is very accurate medical diagnosis using deep learning, and one such successful application is the recognition of ALL. Various deep learning techniques, including CNN, ResNet-50, Xception, MobileNetV2, and VGG19, have been employed on image datasets to tackle this critical issue. Experimental results have demonstrated that CNN, Xception, and MobileNet achieve remarkable 100% accuracy, outperforming other models. The proposed system holds significant potential in the healthcare domain, offering efficient diagnostic assistance to combat the severity of ALL. In the future, even higher accuracies can be achieved by exploring different learning methods and activation functions. The software can be optimized in future to enable early detection of Acute-Lymphoblastic-Leukemia, leading to improved patient recovery and personalized treatment recommendations. Future efforts will focus on refining the Acute-Lymphoblastic- Leukemia extraction process based on current results, giving more thorough explanations of the problem. Additionally, the system can be extended to

detect other subtypes of Leukemia, broadening its scope and impact on healthcare. These advancements will further bolster the system's capabilities and contribute to the ongoing fight against Acute Lymphoblastic Leukemia, ultimately saving more lives and enhancing medical outcomes.

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